higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 3

20 NOVEMBER 2019

This marking guideline consists of 5 pages.

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SECTION A

QUESTION 1

1.1 D
1.2 D
1.3 C
1.4 D
1.5 B

(5 × 2) **[10]**

QUESTION 2

2.1 H
2.2 C
2.3 I
2.4 B
2.5 G
2.6 D
2.7 E
2.8 F

(8 × 2) **[16]**

TOTAL SECTION A: 26

SECTION B

QUESTION 3

3.1
- Those based on the height of a liquid column
- Those that measure by how much an elastic chamber has collapsed
- Those that measure the pressure electrically (3 × 2) (6)

3.2
- It can be used to view the flow.
- It can be used to view the colour.
- It can be used to check the quality of the liquid.
- It can be used to check whether the tank is overflowing or empty.

(Any 2 × 1) (2)

3.3 A float can easily be attached to a measuring device outside the tank. The needle or pointer on the outside moves with the float on the surface, indicating the liquid level. (3)

3.4
- Diaphragm gauges
- Bellows-type of gauges
- Bourdon tube
- Capsule-element gauges (4)

3.5
- Magnetism
- Resistance
- Capacitance
- Piezo-electric (Any 3 × 1) (3)

3.6 It is a measurement of the average kinetic energy of molecules in an object or system or the degree of hotness or coldness of a body. (2)
[20]

QUESTION 4

4.1 4.1.1 A lower control limit is the bottom limit in control for data points that fall below the average line in the control chart.

4.1.2 A normal control limit is the central line on the control chart and the preferred measured value.

4.1.3 A higher control limit is the top limit above the average line in a control chart.

(3 × 2) (6)

4.2
- Pressure instruments
- Temperature instruments
- Flow instrument
- Level instrument (4)

4.3 System errors are caused by a fault within the system element.✓✓
A parallax error occurs when you make an incorrect reading on the scale by not looking at the pointer perpendicular to the reading.✓✓ (2 × 2) (4)

4.4 A distributed control system is a control system which is used in processes✓ in which the controller elements are spread out through the system instead of being localised.✓ (2)
[16]

QUESTION 5

5.1 5.1.1 An indicator is a manually readable device that displays the process information sent to it by the transmitter.

5.1.2 A recorder registers measurements to make available a continuous, permanent record of the values of the variable.

5.1.3 A controller examines measurements and determines what action should be taken.

(3 × 2) (6)

5.2 Legends are used on PFDs to describe symbols and line structures used on that specific PFD. (2)

5.3 5.3.1



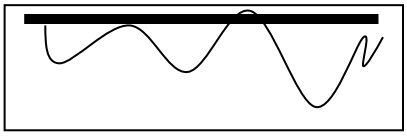
✓✓

5.3.2



✓✓

5.3.3



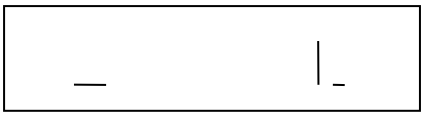
✓✓

5.3.4



✓✓

5.3.5



✓✓

(5 × 2) (10)
[18]

QUESTION 6

6.1 Process control loops control the manufacturing process✓✓ and influence the quality of products while they are being produced. This involves collecting data on the process or product, comparing it to the set point✓✓ and using this information to decide if the process needs to be corrected.✓✓ (6)

6.2
- In-line blending
- Batch blending (2)

6.3 A – Disturbance
B – Feed forward
C – Sensor
D – Demand set point
E – Controller
F – System
G – Output

(7 × 1) (7)
[15]

QUESTION 7

7.1 True
7.2 True
7.3 True
7.4 False
7.5 True

(5 × 1) **[5]**

TOTAL SECTION B: 74
GRAND TOTAL: 100